

2021-Ph-H1-Q2

Section: Our Dynamic Universe

Topic: Motion, Equations and Graphs

You're shown a velocity–time graph and asked to select the correct acceleration–time graph.

Solution:

Acceleration is the **gradient** of a velocity–time graph.

If $v-t$ is a straight sloped line, acceleration is constant.

If $v-t$ is horizontal, acceleration is zero.

If $v-t$ slopes downward, acceleration is negative and constant.

Answer: C

Guidance for Students:

Match the slope of each $v-t$ segment to its corresponding acceleration. Don't confuse speed with acceleration.

Revision Tip:

- $v-t$ gradient = acceleration
- $a-t$ area = change in velocity

Be fluent in translating between graphs.